Theory of Mechanism and Machine

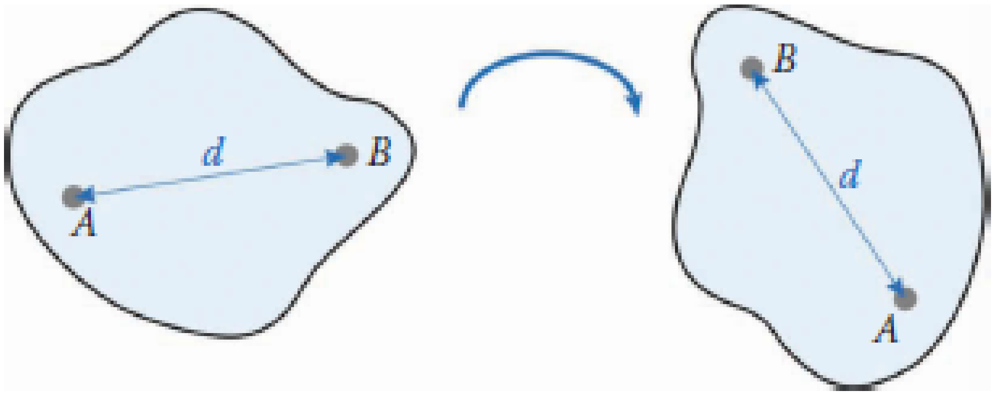
(Week-1)

INTRODUCTION:

The theory of machines and mechanisms is an applied science that is used to understand the relationships between the geometry and motions of the parts of a machine, or mechanism, and the forces that produce these motions.

- Kinematics is the study of motion, quite apart from the forces that produce the motion. In particular, kinematics is the study of position, displacement, rotation, speed, velocity, acceleration, and jerk.
- The study, say, of planetary or orbital motion is also a problem in kinematics, but in this course we shall concentrate our attention on kinematic problems that arise in the design and operation of mechanical systems.

- The first concept is that of a *rigid body. In a rigid body, any two points that are separated by a distance 'd' maintain that distance regardless of the ensuing motion. In other words, a rigid* body cannot stretch, twist, compress, or otherwise deform.



Kinematics It deals with the relative motions of different parts of a mechanism without taking into consideration the forces producing the motions. Thus, it is the study, from a geometric point of view, to know the displacement, velocity and acceleration of a part of a mechanism.

Dynamics It involves the calculations of forces impressed upon different parts of a mechanism. The forces can be either static or dynamic. Dynamics is further subdivided into *kinetics* and *statics*. Kinetics is the study of forces when the body is in motion whereas statics deals with forces when the body is stationary.

Link:

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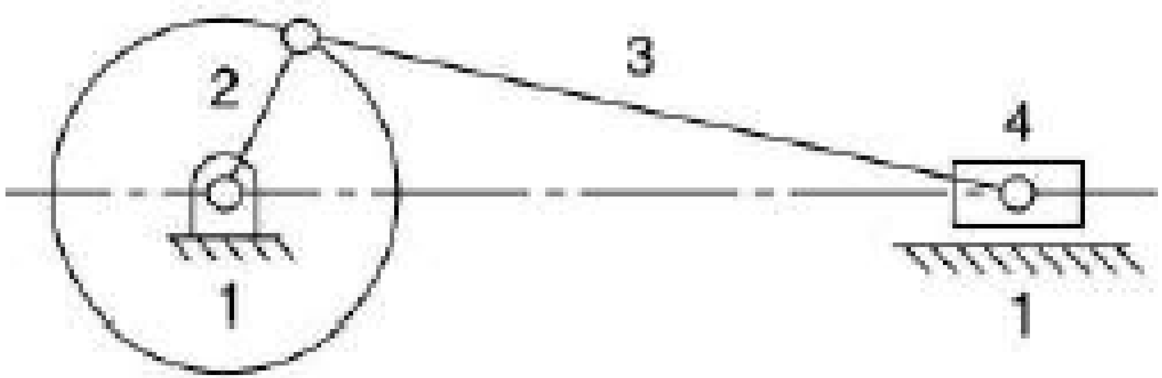
Links can be also classified into binary, ternary and quaternary depending upon their end on which revolute or turning pairs can be placed.

Kinematic Pair:

A kinematic pair is a joint of two links having relative motion between them.

As shown in the below figure, in a slider crank mechanism, the link-2 rotates relative to the link-1 and constitutes a revolute or turning pair. Similarly, links-2,3&4 constitute turning pairs. Link-4 (slider) reciprocates relative to the link-1 and is a sliding pair.



Types of Kinematic Pairs Kinematic pairs can be classified according to

- nature of contact
- nature of mechanical constraint
- nature of relative motion

Kinematic Pairs according to Nature of Contact

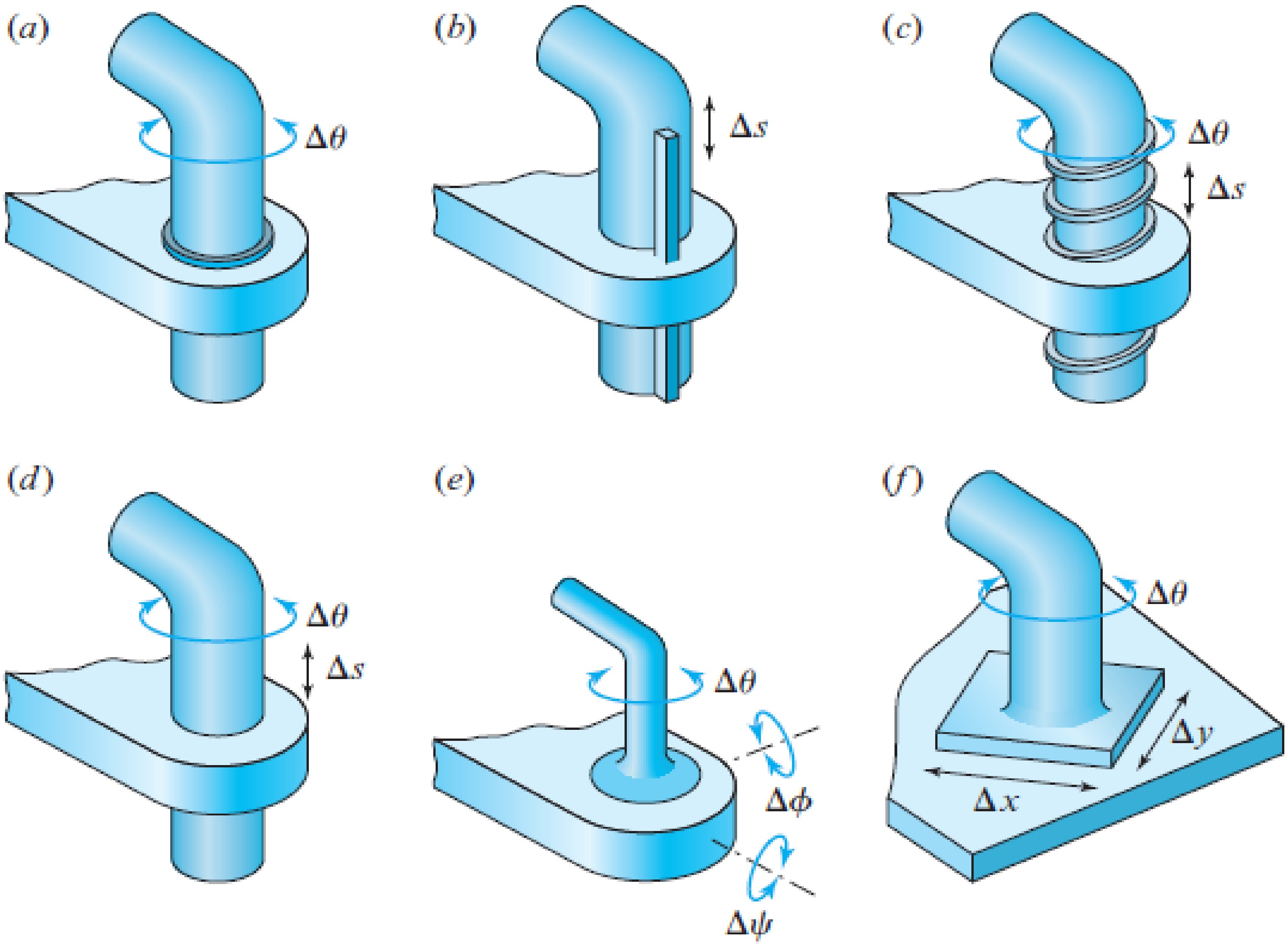
(a) Lower Pair A pair of links having surface or area contact between the members is known as a lower pair. The contact surfaces of the two links are similar.

Examples Nut turning on a screw, shaft rotating in a bearing, all pairs of a slider-crank mechanism, universal joint, etc.

(b) Higher Pair When a pair has a point or line contact between the links, it is known as a higher pair. The contact surfaces of the two links are dissimilar.

Examples Wheel rolling on a surface, cam and follower pair, tooth gears, ball and roller bearings, etc.

Lower pairs consist of the six prescribed types as shown in Fig-



(a) Revolute; (b) prism; (c) screw; (d) cylinder; (e) sphere; (f) flat pairs.

Table 1.1 Lower Pairs

Pair	Symbol	Pair Variable	Degrees of Freedom	Relative Motion
Revolute	R	$\Delta\theta$	1	Circular
Prism	P	Δs	1	Rectilinear
Screw	H	$\Delta\theta$ or Δs	1	Helical
Cylinder	C	$\Delta\theta$ and Δs	2	Cylindric
Sphere	S	$\Delta\theta,\ \Delta\phi,\ \Delta\psi$	3	Spheric
Flat	F	$\Delta x, \Delta y, \Delta\theta$	3	Planar

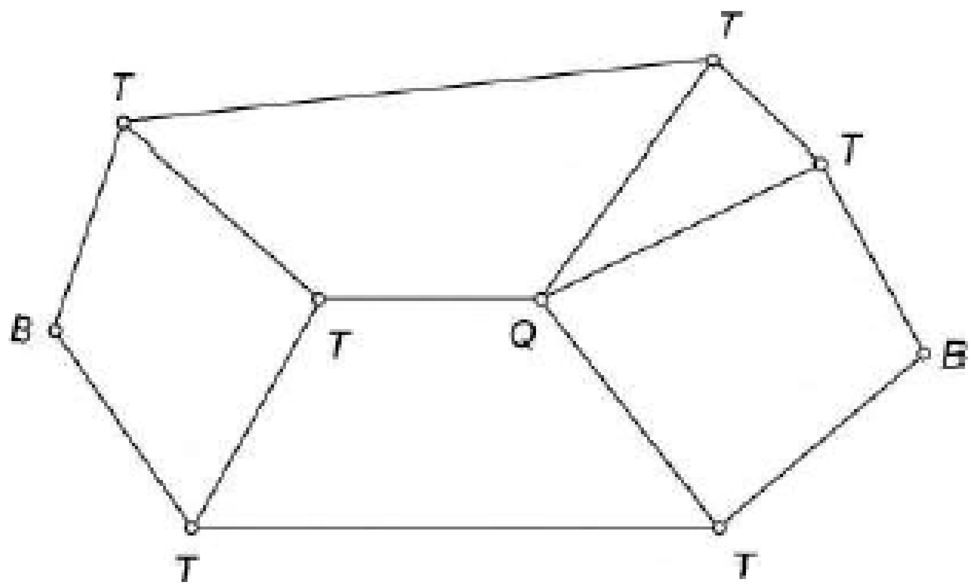
Types of joints: The usual types of joints in a chain are binary, ternary and quaternary joints.

Binary Joint: If two links are joined at the same connection, it is called a binary joint.

Ternary Joint: If three links are joined at the same connection, it is called a binary joint. It is considered equivalent to two binary joints.

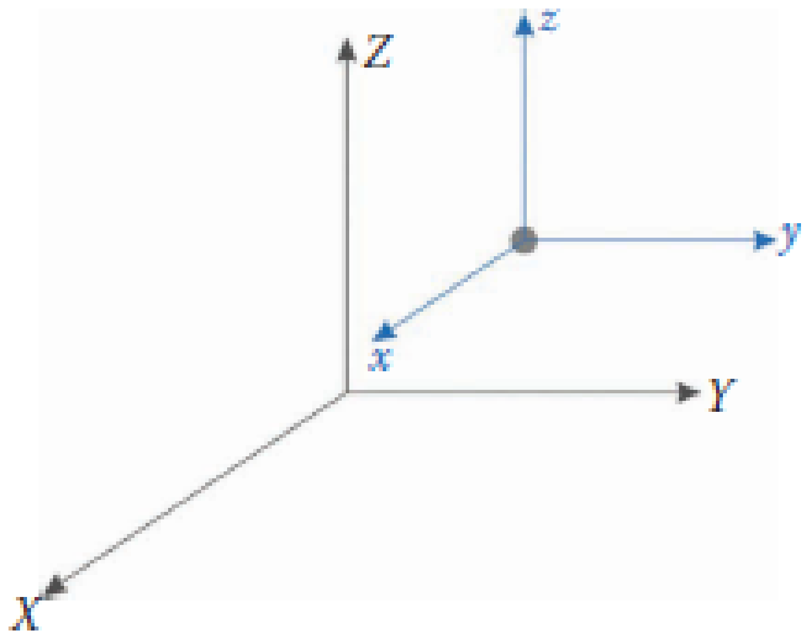
Quaternary Joint: If four links are joined at the same connection, it is called a binary joint. It is considered equivalent to three binary joints.

If 'n' numbers of links are connected at a joint, It is equivalent to (n-1) binary joint.



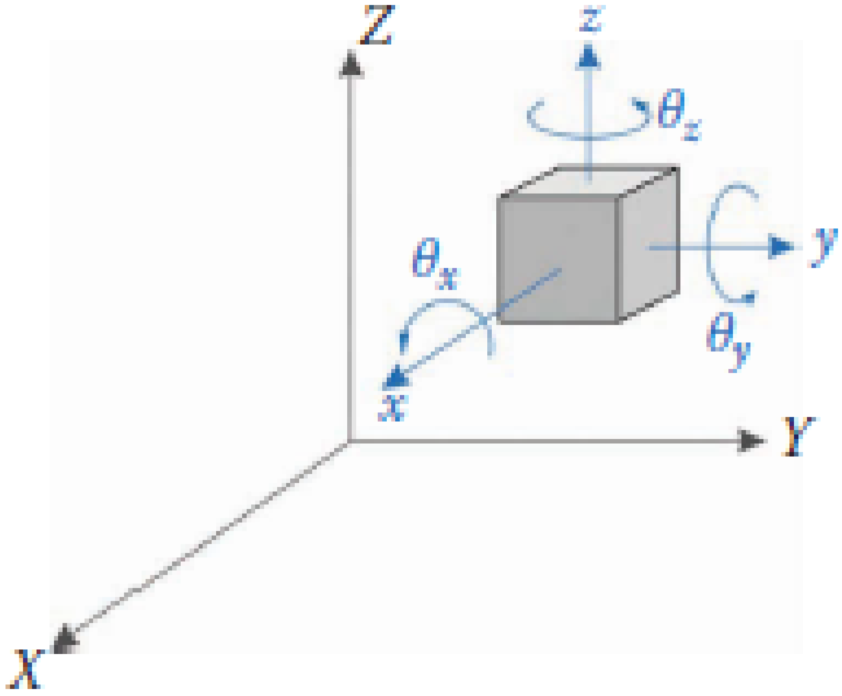
- Degrees of Freedom:
- *In designing a mechanism, it is often critical to know its mobility, or the number of degrees of freedom (DOF) it possesses.*
- *To determine the mobility of a mechanism, we define the number of DOF as:*
- *DOF=Number of independent coordinates needed to completely define an object's orientation in space.*

- *Imagine a point mass in 3D space like the one shown in Figure . The point is free to move in three directions: x, y, and z. It would take three coordinates to completely specify the position of the point mass; therefore, it has three DOF.*



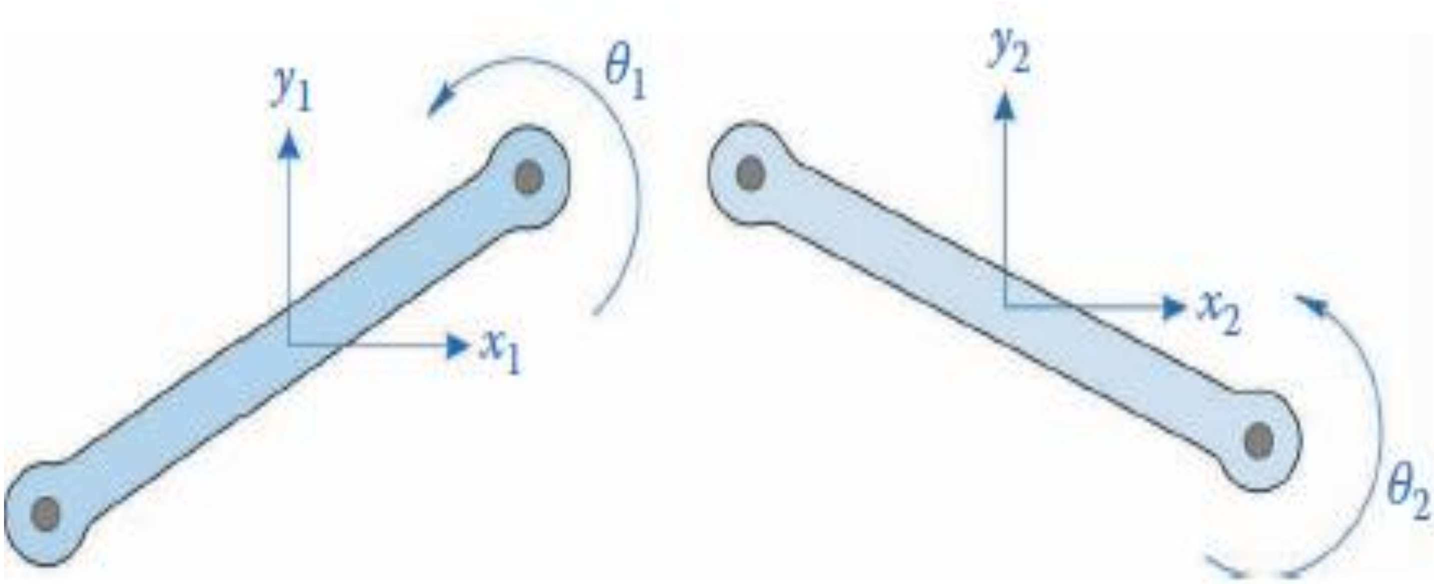
A point in 3D space has three DOF: three translations

- Now imagine a rigid body in 3D space. The body can translate in the same three directions as the point mass, but it can also rotate about its three axes as shown in Figure.
- To specify the configuration of the rigid body requires six coordinates: three translations and three rotations. Thus, a rigid body in 3D space has six DOF.



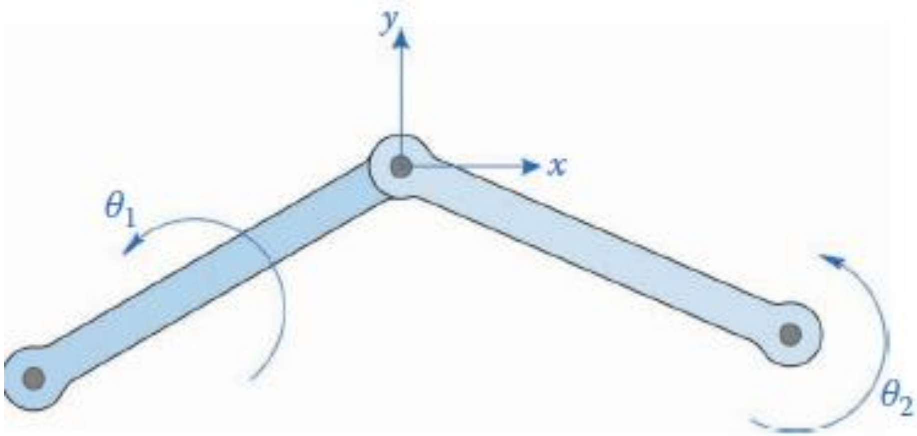
A rigid body in 3D space has six DOF: three translations and three rotations.

- Mobility of Mechanisms:
- *With two bodies or links, we have six DOF, since each body has three of its own . We can generalize this by saying*
- *DOF=3L, where L is the number of links*



Two links have six DOF: four translations and two rotations

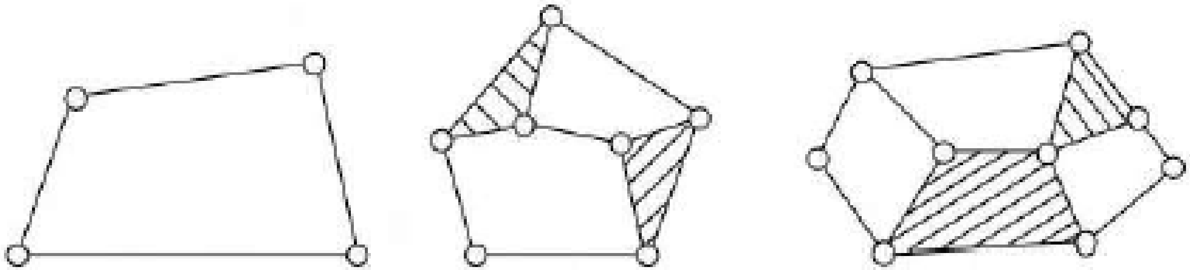
Adding the pin constrains the translation of each link at the location of the pin without restricting rotation of each link. Denote this location (x_1, y_1) on link-1 and (x_2, y_2) on link-2. Then, for the pin joint, one can have:



A pin joint removes two DOF

It now appears that coordinates x_2, y_2 (or x_1, y_1) are not independent; in fact, they have been eliminated as DOF. Thus, adding a pin joint removes two DOF from a mechanism. DOF = 3L-2Jp.
where Jp is the number of pin joints.

Kinematic Chain: A kinematic chain is an assembly of links in which the relative motions of the links is possible and the motion of each relative to the other is definite.



(a) (b) (c)

Linkage Mechanism and Structure:

A linkage is obtained if one of the links of a kinematic chain is fixed to the ground. If motion of any of the moveable links results in definite motions of the others, the linkage is known as a mechanism.

If one of the links of a redundant chain is fixed , it is known as a structure or a locked system.

To obtain constrained or definite motions of some of the links of a linkage , it is necessary to know how many inputs are needed.

The degree of freedom of a structure or a locked system is zero. A structure with negative DOF is known as a superstructure.

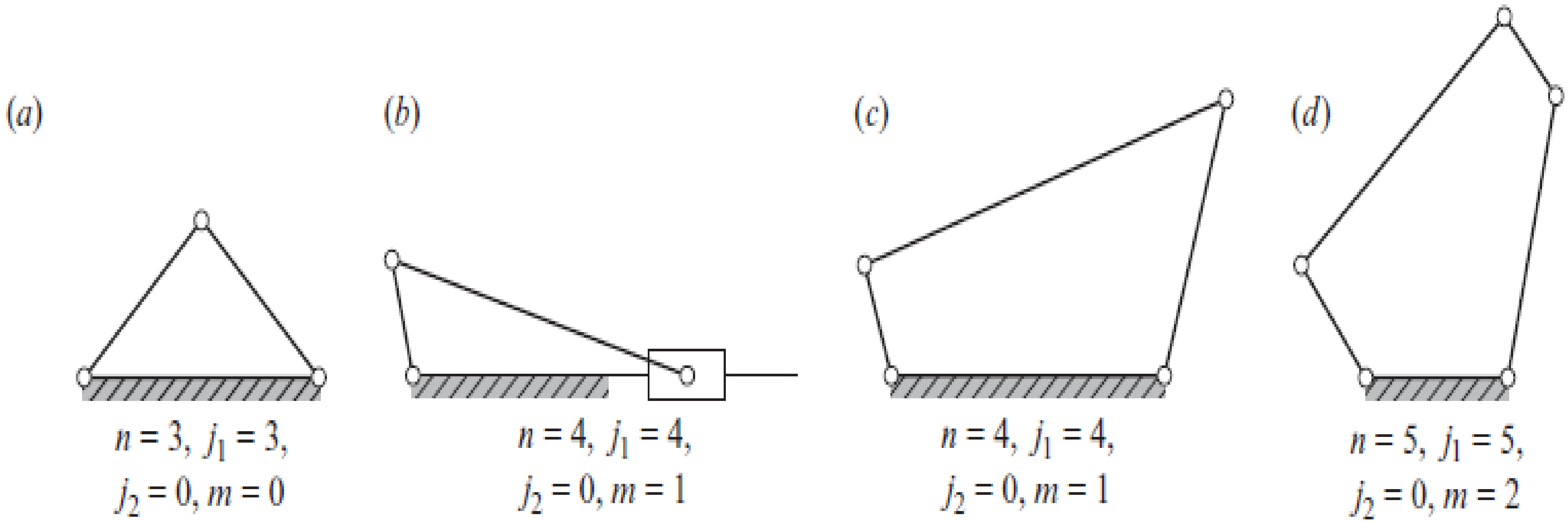
Mobility of Mechanism:

The mobility of a mechanism is the number of input parameters (usually joint variables) that must be controlled independently to bring the device into a particular posture.

If we denote the number of single-degree-of-freedom joints as j1 and the number of two-degree-of-freedom joints as j2, then the resulting mobility (m) of a planar n-link mechanism is given by-

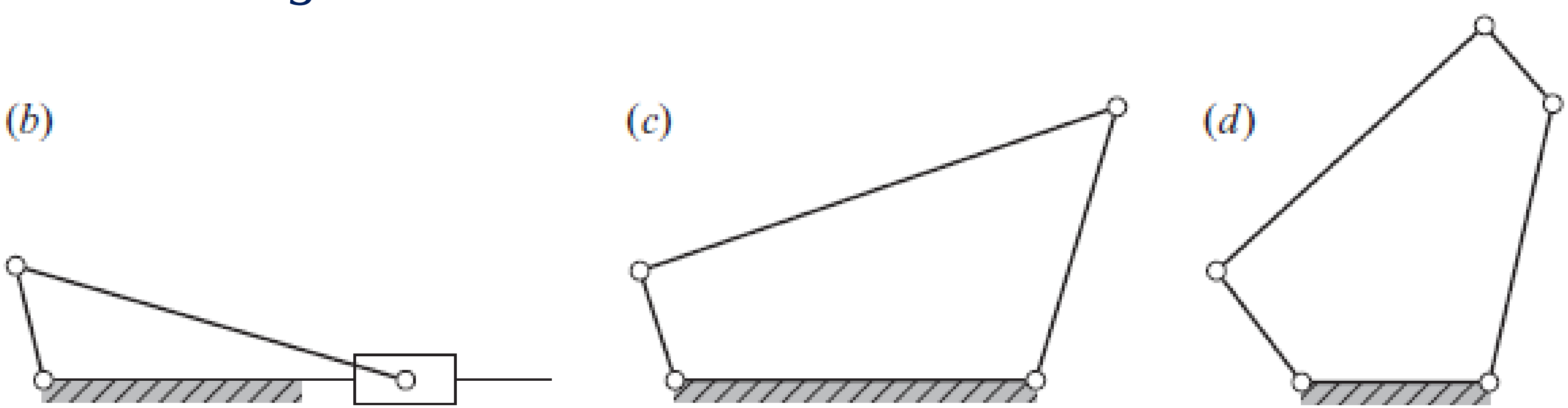
$$m = 3(n-1) - 2j_1 - j_2.$$

This is called the Kutzbach criterion for the mobility of a planar mechanism.



Applications of the Kutzbach criterion.

- *If the Kutzbach criterion yields m>0, the mechanism has m degrees of freedom*.
- *If m=1, the mechanism can be driven by a single input motion to produce constrained (uniquely defined) motion.*
- *Two examples are the slider-crank linkage and the four-bar linkage as shown in Figs. b and c, respectively. If m=2, then two separate input motions are necessary to produce constrained motion for the mechanism; such a case is the five-bar linkage shown in Fig. d.*



(*b*) $n = 4,\ j_1 = 4,\ j_2 = 0,\ m = 1$

(*c*) $n = 4,\ j_1 = 4,\ j_2 = 0,\ m = 1$

(*d*) $n = 5,\ j_1 = 5,\ j_2 = 0,\ m = 2$

Expressing the number of DOF of a linkage in terms of number of links and number of pair connections of different types is known as number synthesis.

The DOF of a mechanism in space can be determined as follows:

N=Total number of links in a mechanism, F=DOF

P1=No of pairs having 1-DOF, P2=No of pairs having 2-DOF

In a mechanism, if one link is fixed, then there will be (N-1) moveable links.

No of DOF for (N-1) links=6(N-1)

Each pair having 1-DOF imposes 5 restraints on the mechanism, reducing DOF by $5P_1$

Each pair having 2-DOF imposes 4 restraints on the mechanism, reducing DOF by $4P_2$.

Similarly others pair having 3,4,5 DOF reduce DOF of the mechanism.

Thus ,

$$F = 6(N-1) - 5P_1 - 4P_2 - 3P_3 - 2P_4 - P_5$$

This is called the Kutzbach criterion for the mobility of a spaital mechanism

For plane mechanism, the following relation may be used to find out the DOF.

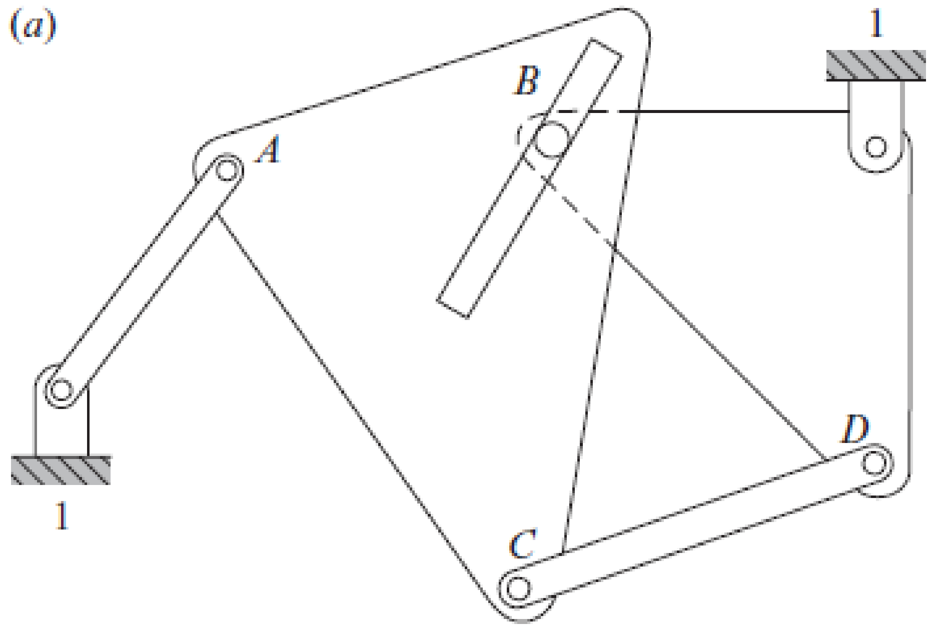
$$F = 3(N-1) - 2P_1 - 1P_2$$

This is known as Grübler criterion for plane mechanism.

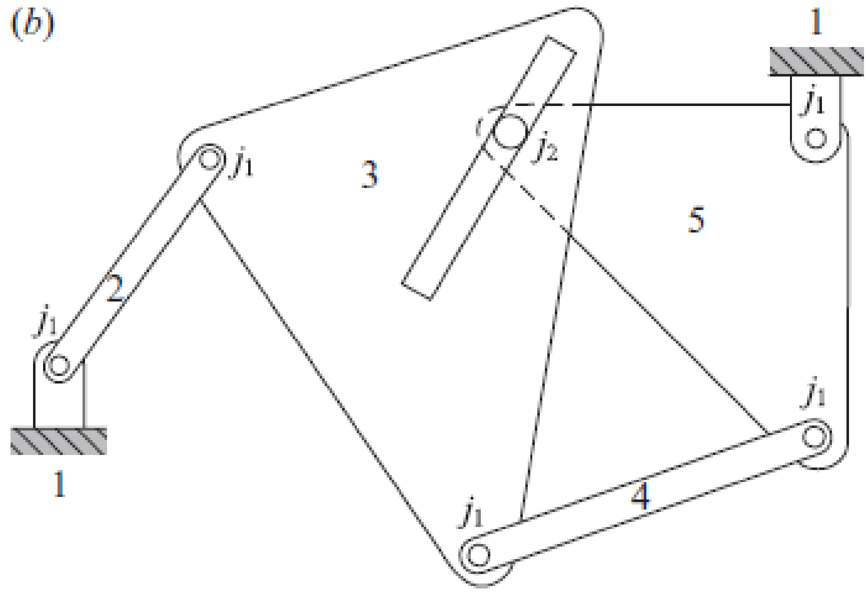
The Grübler criterion for spatial linkages is given by:

$$6n - 5j_1 - 7 = 0.$$

Q 1: Determine the mobility of the planar mechanism shown in Fig-a.



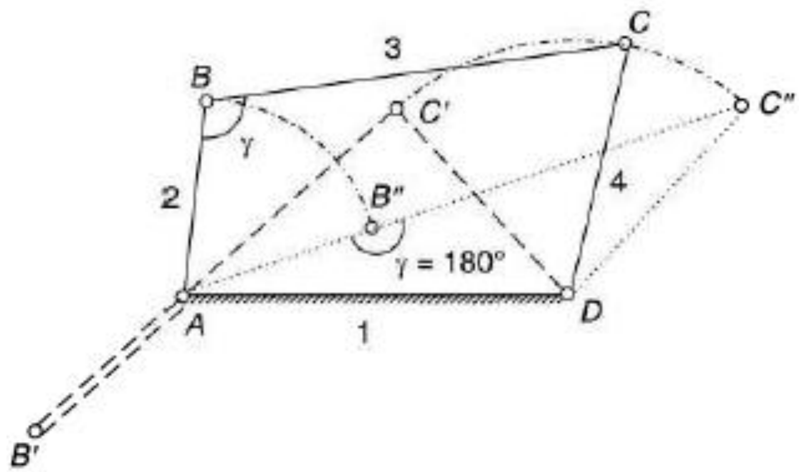
- *The link numbers and the joint types for the mechanism are shown in Fig. b.*



- *The number of links is n=5, the number of lower pairs is j1=5, and the number of higher pairs is j2=1. Substituting these values into the Kutzbach criterion, , the mobility of the mechanism is-*

$$m = 3(5-1)-2(5)-1(1)=1.$$

Mechanical Advantages of a mechanism is the ratio of the output force or torque to the input force or torque at any instant.



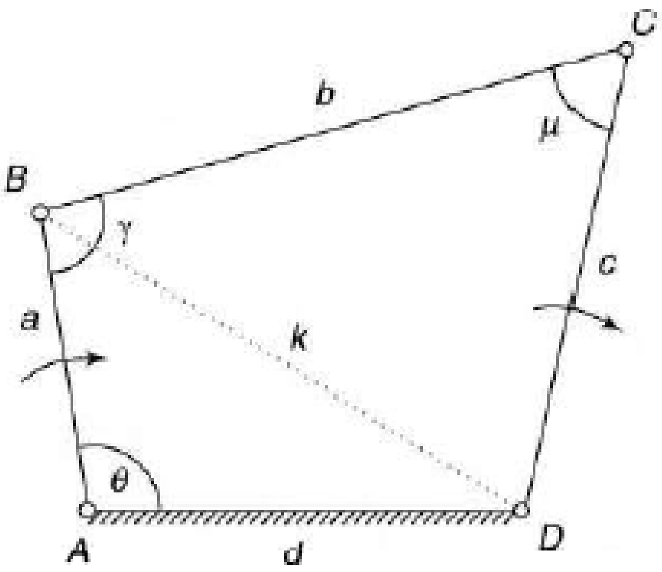
If friction& inertia forces are ignored and the input torque T2 is applied to the link-2 to drive the output link-4 with a resisting torque T4, then

$$\text{Power input} = \text{Power output}$$

$$T_2\,\omega_2 = T_4\,\omega_4$$

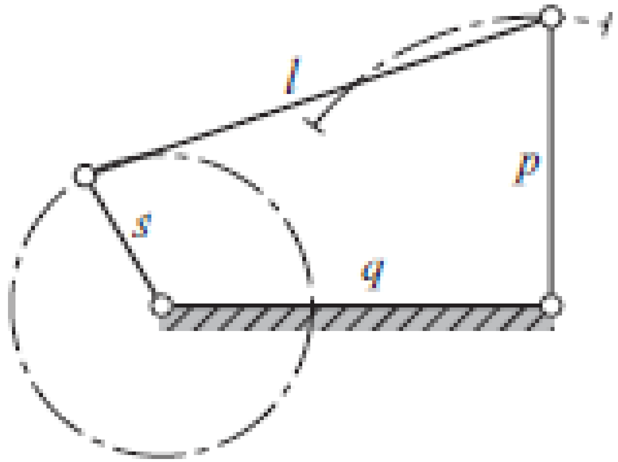
$$\text{or} \qquad \text{MA} = \frac{T_4}{T_2} = \frac{\omega_2}{\omega_4}$$

Transmission Angle: The angle ‘μ’ between the output link and coupler is known as transmission angle.



Grashof's Law:

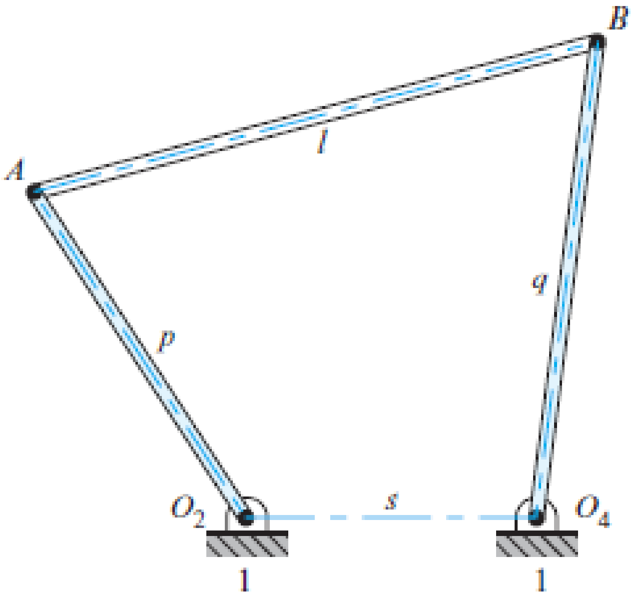
- *Grashof's law states that, for a planar four-bar linkage, the sum of the shortest and longest link lengths cannot be greater than the sum of the remaining two link lengths if there is to be continuous relative rotation between two links.*



- *Where the longest link has length 'l', the shortest link has length 's', and the other two links have lengths 'p' and 'q'. The shortest link will rotate continuously relative to the other three links if and only if*

$$s + l \leq p + q.$$

Q 2: Find the length of link ‘q’ to satisfy a four bar linkage.



p=4 in, l=6in, and s=3in

$$6+3 \leq 4+q$$

Then $5 \leq q$